

1. (a) $xy' - 2y = -x \Rightarrow y' - \frac{2}{x}y = -1$
 $H(x) = \int -\frac{2}{x}dx$, the integrating factor is x^{-2}
 So $x^{-2}y' - \frac{2}{x^3}y = -x^{-2} \Rightarrow \frac{d}{dx}(x^{-2}y) = x^{-2} \Rightarrow x^{-2}y = \frac{1}{x} + C \Rightarrow y = x + Cx^2$
- (b) $y' = \frac{1+2y^2}{y \sin x} \Rightarrow \frac{y}{1+2y^2}dy = \csc x dx \Rightarrow \frac{1}{4} \ln(1+2y^2) = \ln|\csc x - \cot x| + C$
 So the integral curves can be written as $\ln(1+2y^2) = \ln[C(\csc x - \cot x)^4]$, or as
 $y^2 = K(\csc x - \cot x)^4 - \frac{1}{2}$
- (c) $xy' - y = 2x \ln x \Rightarrow y' - \frac{1}{x}y = 2 \ln x$
 $H(x) = \int -\frac{1}{x}dx$, the integrating factor is $\frac{1}{x}$
 So $\frac{1}{x}y' - \frac{1}{x^2}y = \frac{2}{x} \ln x \Rightarrow \frac{d}{dx}(\frac{1}{x}y) = \frac{2}{x} \ln x \Rightarrow \frac{1}{x}y = (\ln x)^2 + C \Rightarrow y = x(\ln x)^2 + Cx$
- (d) $xy' + 2y = \frac{\cos x}{x} \Rightarrow y' + \frac{2}{x}y = \frac{\cos x}{x^2}$
 $H(x) = \int \frac{2}{x}dx = 2 \ln|x|$, the integrating factor is x^2
 So $x^2y' + 2xy = \cos x \Rightarrow \frac{d}{dx}(x^2y) = \cos x \Rightarrow x^2y = \sin x + C \Rightarrow y = \frac{\sin x}{x^2} + \frac{C}{x^2}$
- (e) $y' + 2xy = xe^{-x^2}$
 $H(x) = \int 2xdx = x^2$, the integrating factor is e^{x^2}
 $e^{x^2}y' + 2xe^{x^2}y = x \Rightarrow \frac{d}{dx}(e^{x^2}y) = x \Rightarrow e^{x^2}y = \frac{1}{2}x^2 + C \Rightarrow y = e^{-x^2}(\frac{1}{2}x^2 + C)$
- (f) $y' = xy^2 - x - y^2 + 1 = (x-1)(y^2-1)$
 $\Rightarrow \int \frac{dy}{y^2-1} = \int \frac{dx}{x-1} \Rightarrow \frac{1}{2} \ln \left| \frac{y-1}{y+1} \right| = \ln|x-1| + C$
 The solution can also be written as $y = \frac{1 + Ke^{x^2-2x}}{1 - Ke^{x^2-2x}}$, where $K = e^C$
- (g) $e^y \sin 2x dx + \cos x(e^{2y} - y)dy = 0$
 $\int \frac{\sin 2x}{\cos x} dx + \int (e^y - ye^{-y})dy = C \Rightarrow -2 \cos x + e^y + e^{-y}(1+y) = C$
- (h) $y' = x\sqrt{\frac{1-y^2}{1-x^2}} \Rightarrow \frac{1}{\sqrt{1-y^2}}dy = \frac{x}{\sqrt{1-x^2}}dx \Rightarrow \int \frac{1}{\sqrt{1-y^2}}dy = \int \frac{x}{\sqrt{1-x^2}}dx \Rightarrow$
 $\arcsin y = -\sqrt{1-x^2} + C$
 Since $y(0) = 0 \Rightarrow \arcsin 0 = -1 + C \Rightarrow C = 1$
 Thus, $\arcsin y = 1 - \sqrt{1-x^2}$
2. Set $z = y' - y$. Then $z' = y'' - y'$
 So $y' - y = y'' - y \Rightarrow z = z' \Rightarrow z = C_1 e^x$
 Now $z = y' - y = C_1 e^x \Rightarrow e^{-x}y' - e^{-x}y = C_1 \Rightarrow (e^{-x}y)' = C_1 \Rightarrow e^{-x}y = C_1 x + C_2$
 $\Rightarrow y = C_1 x e^x + C_2 e^x$

3. (a) $\frac{dP}{dt} + (b-a)P = 0$ $H(t) = \int (b-a)dt = (b-a)t$, the integrating factor is $e^{(b-a)t}$
 $e^{(b-a)t} \frac{dP}{dt} + (b-a)e^{(b-a)t}P = 0 \Rightarrow \frac{d}{dt}(e^{(b-a)t}P) = 0 \Rightarrow e^{(b-a)t}P = C \Rightarrow P = Ce^{(b-a)t}$
- (b) i. $a > b \Rightarrow P_0 e^{(b-a)t}$ is increasing.
 $P(t) \rightarrow \infty$ as $t \rightarrow \infty$
- ii. $a = b \Rightarrow P(t) = P_0$ is a constant.
- iii. $a < b \Rightarrow P_0 e^{(b-a)t}$ is decreasing.
 $P(t) \rightarrow 0$ as $t \rightarrow \infty$
4. (a) $\frac{dP}{dt} = k(M-P)$
- (b) $\frac{dP}{dt} + kP = kM$
 $H(t) = \int kdt = kt$, the integrating factor is e^{kt}
 So $e^{kt} \frac{dP}{dt} + ke^{kt}P = kMe^{kt} \Rightarrow \frac{d}{dt}(e^{kt}P) = kMe^{kt} \Rightarrow e^{kt}P = Me^{kt} + C \Rightarrow P = M + Ce^{-kt}$
 $P(0) = M + C = 0 \Rightarrow C = -M$ and $P(t) = M(1 - e^{-kt})$
 $P(10) = M(1 - e^{-10k}) = 0.3M \Rightarrow k \approx 0.0357$ and $P(t) = M(1 - e^{-0.0357t})$
- (c) $P(t) = M(1 - e^{-0.0357t}) = 0.9M \Rightarrow e^{-0.0357t} = 0.1 \Rightarrow t \approx 65$
 Therefore, it will take approximately 65 days for 90% of population to be aware of the product.
5. (a) Let $P = P(t)$ denote the number of people who have the disease at time t . Then with $M = 25000$ and $R = 100$, we get
- $$P(t) = \frac{25000(100)}{100 + (24900)e^{-25000kt}} = \frac{25000}{1 + 249e^{-25000kt}}$$
- $$P(10) = \frac{25000}{1 + 249e^{-25000(10k)}} = 400 \Rightarrow -25000k \approx -0.1398$$
- Therefore, $P(t) = \frac{25000}{1 + 249e^{-0.1398t}}$
- (b) $\frac{25000}{1 + 249e^{-0.1398t}} = 12500 \Rightarrow t \approx 40$
 It will take 40 days for half the population to have the disease.

